

IS5743: INFORMATION TECHNOLOGY BASED BUSINESS TRANSFORMATION

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Information Technology Based Business Transformation

Subject Code

IS - Information Systems

Course Number

5743

Academic Unit

Information Systems (IS)

College/School

College of Business (CB)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

IS5313 Foundations of Information and Electronic Business Systems

Equivalent Courses

IS6643 Information Technology Based Organization Transformation

Exclusive Courses

Nil

Part II Course Details

Abstract

This course aims to:

- provide students with knowledge and experience relevant to business transformation and business process management, and
- enable students to apply information technology (IT) to enable and facilitate business transformation.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Describe the key essence of process-oriented organizations and the evolution of management philosophy related to business transformation.	25			
2	Apply a disciplined business process management methodology and propose IT-based solutions.	30	x	x	
3	Deploy and implement technology and enterprise systems to enable business transformation.	20			
4	Manage changes due to IT-based business transformation and identify the key success factors of business transformation.	25	x	x	

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Seminar	Students will learn different topics such as environmental analyses for business transformation, business process management in the digital age, strategy and BPM, design thinking, process-oriented organizations, business process reengineering (BPR), the role of IT in business transformation, enterprise resource planning (ERP), enterprise systems, evolution of management philosophy in business transformation, the Leavitt Diamond model of organization adaptation, business process modelling, organization change management, and critical success factors for business transformation.	1, 2, 3, 4	
2	Case Studies	Throughout the semester, students will apply and integrate what they learn through classroom discussions and readings into real-world mini-cases. Students are expected to participate actively in in-class discussions to gain an in-depth understanding of the key issues related to the cases.	2, 3, 4	
3	Project	Students will conduct group projects to apply the concepts learned in class, and use creative and critical thinking skills to communicate the ideas and results of their work.	1, 2, 4	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Continuous Assessment and Self-Reflection The instructor encourages a two-way, interactive learning environment, thus students are expected to participate actively in class. The participation grade will be based on both in-class and online reflective journals (Canvas) participation.	1, 2, 3, 4	20	-	Yes
2	Project Students will work in teams on a project. Students can either: a) Analyse an existing organization, business and processes, then suggest an IT-based transformation solution, or b) Critique a transformation project of an organization that uses information technology to bring advantages to the organization.	1, 2, 4	30	GenAI may be used to generate ideas during preparation, but you are not allowed to copy and paste its output directly into your final submission. If you use GenAI, include your prompts and your own original work in the appendix, and clearly refer to them in your report.	Yes

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

50

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

Note: Students must pass BOTH coursework and examination in order to get an overall pass in this course.

Assessment Rubrics (AR)

Assessment Task

Continuous Assessment (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILOs 1, 2, 3, 4] Participation in Seminars/ Case Discussions

- Attribute to contribute to class discussion actively by asking excellent questions and responding to others' questions frequently and proactively
- Ability to demonstrate the ability to solve many real problems by applying knowledge learned in the class
- Ability to contribute to case discussions proactively and demonstrate a deep understanding of the key issues involved in the cases

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILO 1] Introduction and Background (business environment, and justification of the choice of the process to be transformed)

- Ability to demonstrate a strong justification of the need for business transformation based on environmental analyses
- Ability to provide rich, relevant material based on information from multiple sources
- Ability to use well-structured and grammatically correct description

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILO 2] Redesign of a Business Process

- Ability to appropriately apply the 5-phases model to redesign the process
- Ability to clearly describe the scope of the business process to be redesigned
- Ability to adequately and appropriately apply the redesign principles and tactics
- Ability to propose a well-justified, innovative redesign solution
- Ability to propose alternative redesigns which are considered and ruled out convincingly

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILO 4] Implementation

- Ability to propose actionable, most suitable suggestions to facilitate the implementation of the project
- Ability to identify the most relevant, key success factors of the project

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILO 1] Key concepts and management approach to business transformation

- Ability to effectively distinguish between radical redesign and incremental improvement of business processes ;
- Ability to effectively identify needs for organization transformation based on an environmental analysis
- Ability to demonstrate evidence of deep understanding of the essence of business processes

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILO 2, 3] Design a new business process or redesign an existing business process

- Ability to effectively identify organizational needs for business transformation based on an internal analysis
- Ability to identify the most suitable team members for the transformation project
- Ability to set up the most appropriate goals and objectives for the transformation project
- Ability to appropriately model a business process
- Ability to effectively identify new processes or the key weaknesses of an existing business process
- Ability to effectively identify the principles and tactics that are most suitable for a transformation project and appropriately apply them
- Ability to propose a very suitable IT-based solution to achieve the transformation objectives

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Final Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

[CILO 4] Implementation of organization transformation

- Ability to propose actionable, most suitable advice to facilitate the organization transformation by various models
- Ability to effectively identify the key success factors of a process transformation project

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Continuous Assessment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILOs 1, 2, 3, 4] Participation in Seminars/ Case Discussions

- Attribute to contribute to class discussion actively by asking excellent questions and responding to others' questions frequently and proactively
- Ability to demonstrate the ability to solve many real problems by applying knowledge learned in the class
- Ability to contribute to case discussions proactively and demonstrate a deep understanding of the key issues involved in the cases

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILO 1] Introduction and Background (business environment, and justification of the choice of the process to be transformed)

- Ability to demonstrate a strong justification of the need for business transformation based on environmental analyses
- Ability to provide rich, relevant material based on information from multiple sources
- Ability to use well-structured and grammatically correct description

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILO 2] Redesign of a Business Process

- Ability to appropriately apply the 5-phases model to redesign the process
- Ability to clearly describe the scope of the business process to be redesigned
- Ability to adequately and appropriately apply the redesign principles and tactics
- Ability to propose a well-justified, innovative redesign solution
- Ability to propose alternative redesigns which are considered and ruled out convincingly

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILO 4] Implementation

- Ability to propose actionable, most suitable suggestions to facilitate the implementation of the project
- Ability to identify the most relevant, key success factors of the project

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Final Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILO 1] Key concepts and management approach to business transformation

- Ability to effectively distinguish between radical redesign and incremental improvement of business processes
- Ability to effectively identify needs for organization transformation based on an environmental analysis
- Ability to demonstrate evidence of deep understanding of the essence of business processes

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Final Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILO 2, 3] Design a new business process or redesign an existing business process

- Ability to effectively identify organizational needs for business transformation based on an internal analysis
- Ability to identify the most suitable team members for the transformation project
- Ability to set up the most appropriate goals and objectives for the transformation project
- Ability to appropriately model a business process
- Ability to effectively identify new processes or the key weaknesses of an existing business process
- Ability to effectively identify the principles and tactics that are most suitable for a transformation project and appropriately apply them
- Ability to propose a very suitable IT-based solution to achieve the transformation objectives

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Final Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

[CILO 4] Implementation of organization transformation

- Ability to propose actionable, most suitable advice to facilitate the organization transformation by various models
- Ability to effectively identify the key success factors of a process transformation project

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information**Keyword Syllabus**

Business Transformation, Digital Transformation, Environmental Analyses for Business Transformation, Business Process Management, Design Thinking, Alignment of Business Model and Processes, the Role of IT in Business Transformation, Evolution of Management Philosophy in Business Transformation, Artificial Intelligence in BPM, Workflow Management, Enterprise Systems, the Leavitt Diamond Model; BPR Methodology; IT Integration; Change Management; and Critical Success Factors.

Reading List**Compulsory Readings**

Title	
1	Nil

Additional Readings

Title	
1	Lars Reinkemeyer, "Process Mining in Action: Principles, Use Cases and Outlook", 1st edition, Springer, 2020.
2	Jace An, "77 Building Blocks of Digital Transformation: The Digital Capability Model", Story Tree FDC, 2019.
3	David L, Rogers "The Digital Transformation Playbook : Rethink Your Business for the Digital Age", Columbia University Press 2016.
4	Jan vom Brocke and Theresa Schmiedel, "BPM - Driving Innovation in a Digital World", Springer, 2015.
5	El Sawy, O.A., "Redesigning Enterprise Processes for e-Business" McGraw-Hill, 2001.